

Mathematics A

Unit 1 • Chapter OL-T09 Classified

Cross-session • chapter-organised candidate

22 classified items • 63 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Unit 1 • Chapter OL-T09 • 22 items • 63 marks

Chapter	Items	Marks	Starts
01. OL-T09 Rounding, Estimation & Bounds	22	63	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Rounding, Estimation & Bounds

Unit 1 • 63 marks • 22 item(s)

June 2025 | 4MA1-1H Q2 | 2 marks

Direct lower and upper bounds

June 2025 | 4MA1-2H Q22 | 4 marks

Geometry, rate or practical decision bounds

June 2024 | 4MA1-2H Q19 | 3 marks

Algebraic expression upper or lower bound

November 2023 | 4MA1-2H Q17 | 3 marks

Algebraic expression upper or lower bound

June 2023 | 4MA1-2H Q25 | 4 marks

Geometry, rate or practical decision bounds

November 2024 | 4MA1-1H Q21 | 3 marks

Algebraic expression upper or lower bound

November 2024 | 4MA1-2H Q2 | 1 marks

Direct lower and upper bounds

November 2024 | 4MA1-2H Q2 | 1 marks

Direct lower and upper bounds

June 2022 | 4MA1-2H Q18 | 3 marks

Algebraic expression upper or lower bound

January 2022 | 4MA1-1H Q18 | 3 marks

Algebraic expression upper or lower bound

January 2022 | 4MA1-2H Q4 | 1 marks

Direct lower and upper bounds

January 2022 | 4MA1-2H Q4 | 1 marks

Direct lower and upper bounds

January 2020 | 4MA1-1H Q16 | 3 marks

Algebraic expression upper or lower bound

January 2021 | 4MA1-1H Q15 | 3 marks

Algebraic expression upper or lower bound

June 2021 | 4MA1-1H Q20 | 3 marks

Algebraic expression upper or lower bound

... 7 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 01

4MA1-1H Q2

UNIT 1**2 marks**

The height of each cup is 9 cm, correct to the nearest cm

(b) Write down the lower bound of the height.

..... cm
(1)

The weight of each cup is 120 g, correct to the nearest 10 g

(c) Write down the upper bound of the weight.

..... g
(1)

WORKING SPACE

Working Unit 24

4MA1-2H Q22

UNIT 1
4 marks

- 22 The diagram shows a square inside rectangle $ABCD$

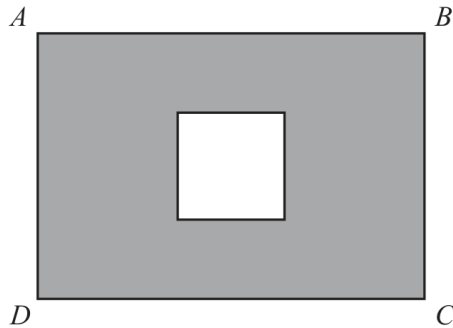


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X \text{ cm}^2$

$AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2 \text{ cm}$ correct to 2 significant figures

side of square = 4.1 cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

WORKING SPACE

Question 26

4MA1-2H Q19

MODULAR UNIT 1

3 marks

19 $G = \frac{c}{2f - 3h}$

$c = 8$ correct to the nearest whole number

$f = 6.62$ correct to 2 decimal places

$h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of G

Give your answer correct to 3 decimal places.

Show your working clearly.

WORKING SPACE

Question 29

4MA1-2H Q17 whole

MODULAR UNIT 1

3 marks

17 $P = a(c + y)$

 $a = 8.3$ correct to 2 significant figures $c = 2$ correct to 1 significant figure $y = 15$ correct to the nearest 5Work out the upper bound for the value of P

Show your working clearly.

WORKING SPACE

Question 29

4MA1-2H Q25

MODULAR UNIT 1

4 marks

- 25 A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.

Give your answer in g/cm^3 correct to 2 decimal places.

Show your working clearly.

..... g/cm^3

WORKING SPACE

Question 16

4MA1-1H Q21 M01

MODULAR UNIT 1

3 marks

21 $T = \frac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.

Show your working clearly.

.....

WORKING SPACE

Question 18

4MA1-2H Q2 M01

MODULAR UNIT 1

1 marks

2 The length of a table is measured as 1.4 metres correct to one decimal place.

(a) Write down the upper bound of the length of the table.

..... metres

(1)

WORKING SPACE

Question 19

4MA1-2H Q2 M02

MODULAR UNIT 1

1 marks

2 The length of a table is measured as 1.4 metres correct to one decimal place.

(b) Write down the lower bound of the length of the table.

..... metres

(1)

WORKING SPACE

Question 25

4MA1-2H Q18 Q18

MODULAR UNIT 1

3 marks

18 $X = \frac{2a - b}{f}$

$a = 7.5$ correct to 1 decimal place.

$b = 3.42$ correct to 2 decimal places.

$f = 2$ correct to the nearest whole number.

Work out the upper bound of the value of X

Show your working clearly.

WORKING SPACE

Question 15

4MA1-1H Q18

MODULAR UNIT 1

3 marks

18 $A = w - \frac{x^2}{y}$

$w = 3.45$ correct to 2 decimal places.

$x = 1.9$ correct to 1 decimal place.

$y = 5$ correct to the nearest whole number.

Work out the lower bound of the value of A
Show your working clearly.

WORKING SPACE

Question 21

4MA1-2H Q4 Q4(a)

MODULAR UNIT 1

1 marks

- 4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

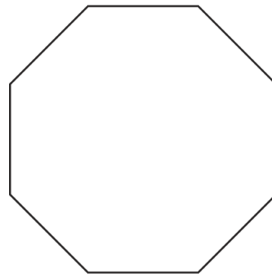


Diagram **NOT**
accurately drawn

- (a) Write down the lower bound of the length of each side of the octagon.

..... mm
(1)

WORKING SPACE

Question 22

4MA1-2H Q4 Q4(b)

MODULAR UNIT 1

1 marks

(b) Write down the upper bound of the length of each side of the octagon.

..... mm

(1)

WORKING SPACE

16 $P = \frac{2a - c}{d}$

$a = 58.4$ correct to 3 significant figures.

$c = 20$ correct to 2 significant figures.

$d = 3.6$ correct to 2 significant figures.

Work out the upper bound for the value of P .

Show your working clearly.

Give your answer correct to 2 decimal places.

YOUR WORKING

15

$$a = \frac{v - u}{t}$$

$v = 9.6$ correct to 1 decimal place

$u = 3.8$ correct to 1 decimal place

$t = 1.84$ correct to 2 decimal places

Calculate the upper bound for the value of a .

Give your answer as a decimal correct to 2 decimal places.

Show your working clearly.

YOUR WORKING

20 $P = \frac{t - w}{y}$

$t = 9.7$ correct to 1 decimal place

$w = 5.9$ correct to 1 decimal place

$y = 3$ correct to 1 significant figure

Calculate the upper bound for the value of P .
Show your working clearly.

YOUR WORKING

13 Here is a triangle XYZ .

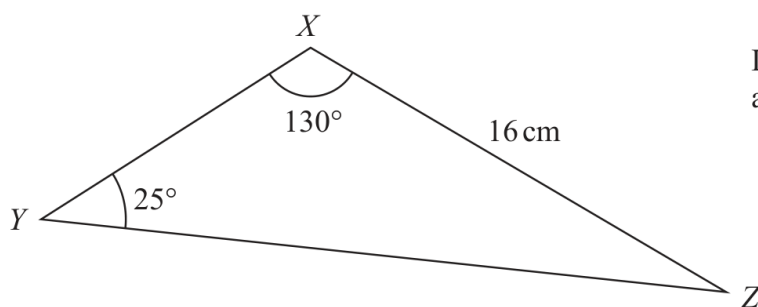


Diagram **NOT**
accurately drawn

The length XZ and the angles YXZ and XYZ are each given correct to 2 significant figures.

Calculate the upper bound for the length YZ .

Give your answer correct to one decimal place.

Show your working clearly.

..... cm

YOUR WORKING

- 17** A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3}) \text{ m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

..... kg/m^3

(Total for Question 17 is 4 marks)

YOUR WORKING

19 The acceleration, a , of an object is given by

$$a = \frac{v - u}{t}$$

where

$v = 45.23$ correct to 2 decimal places

$u = 5.12$ correct to 2 decimal places

$t = 8.5$ correct to 2 significant figures

By considering bounds, work out the value of a to a suitable degree of accuracy.
Show your working clearly and give a reason for your answer.

$a = \dots\dots\dots$

(Total for Question 19 is 5 marks)

YOUR WORKING

5 The weight of a cake is 2.75 kg, correct to 2 decimal places.

(a) Write down the lower bound of the weight of the cake.

..... kg
(1)

(b) Write down the upper bound of the weight of the cake.

..... kg
(1)

Penny has worked out $\frac{81.3 \times 59.2}{1.9^2}$ on her calculator.

Her answer is 13 332.299 17

Penny's answer is not sensible.

(c) By rounding each number to one significant figure, work out a suitable estimate to show that her answer is not sensible.
Show your working clearly.

(2)

(Total for Question 5 is 4 marks)

YOUR WORKING

- 2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

(Total for Question 2 is 2 marks)

YOUR WORKING

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g
(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm
(1)

(Total for Question 7 is 2 marks)

YOUR WORKING

November 2025 | 2H | Question 24 | 5 marks

Chapter OL-T09: Rounding, Estimation & Bounds

24 $D = \frac{n}{p - q}$

$n = 10.3$ correct to 1 decimal place

$p = 7.24$ correct to 2 decimal places

$q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

(Total for Question 24 is 5 marks)

YOUR WORKING
